

Smart Plant Monitoring and Controlling System for Greenhouse Application Using IoT

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Abstract—This work provides an IoT-based plant monitoring system for avoiding over-exploitation of resources, ensuring compact design with easy installation and the best environment for the plants to grow. This type of agricultural monitoring system provides environmental monitoring services that help the crop grow in good shape. The system offers the service of storing a database about the environment and soil information acquired from a wireless sensor network deployed in the planted area. Furthermore, it enables users to track environmental data regarding planted crops in real-time using any Internet-enabled device. Our project introduces automatic sunlight prevention and soil moisture control based on data received from the moisture and temperature sensors. The proposed work is implemented using a microcontroller, programmed accordingly to measure the temperature and moisture using respective sensors. It is well capable of controlling temperature by operating a fan, controlling a shade for blocking excess sunlight and controlling moisture by operating a watering device through a motor. The potential outcome of the project work can be used for greenhouse monitoring and controlling systems.

Keywords—Greenhouse, IoT, Internet-enabled device, Automation.

I. INTRODUCTION

IoT technology was used to bridge the gap between the individuals in the article "Things" and their digital equivalents in data frameworks. It is known as the next-generation network (NGN) of the internet [1]. Agriculture is still conducted in India in a traditional manner, and modern technology is slowly adopted. It is critical for stakeholders to reject conventional agricultural techniques and modernise agriculture through technology since agriculture and related businesses employ around 50% of Indians yet contribute for only 15% of the country's GDP [2-3]. Agriculture's economic contribution to India's GDP is gradually declining as the country's broad-based economic growth slows, despite the fact that a large number of people continue to work in the agricultural sector [4]. As a result, there is a pressing need to modernise the system in order to increase productivity and produce healthy organic food [5]. The Internet of Things (IoT) is a branch of the Internet that combines physical items with the capacity to provide more quick-witted services to the world as more information becomes readily available. Several

application sectors, ranging from green IT and vitality effectiveness of coordination, are presently benefiting from Internet of Things principles [6]. A greenhouse provides basic employment methods to its owner and must be financially feasible for the environment in which it operates [7]. Greenhouses are also considered advanced innovations for Protected Horticulture since they address the major natural aspects of light, temperature, and irrigation [8]. The Internet of Things (IoT) and how to use it for various purposes have recently taken a lot of attention. Numerous articles have recently been published on the exciting subject of the Internet of Things (IoT), which is a technology that is now in its infancy. In 2013, Wenshun et al presented a paper in which Zhuangluo sunlight greenhouse monitoring and early warning service platforms based on cloud computing and IoT are designed and implemented to improve management efficiency and reduce production costs [9]. J. Zhao et al. reported research that analyses the integration of control networks and information networks in the Internet of Things (IoT) in actual agricultural production. They also propose integrating remote monitoring with wireless communications [10]. In literature [11] an adaptive neuro fuzzy inference system is used. Collected data goes to the fuzzy control system and then manipulated and make prediction for optimum values. In literature [12], IoT devices are connected in star topology for access to LPWAN base station for low energy consumption. A low cost low energy consumption system is implemented in literature [13-16] using LinkIt Smart 7688 Duo Microcontroller to store data with inbuilt WiFi module and Thingier io platform for data analyzation. The proposed work developed a prototype suitable for greenhouse environment that not only monitors moisture, temperature and humidity but also takes controlling action as per the requirement. The various controlling action includes control of temperature by operating a fan, control of a shade for blocking excess sunlight and control of moisture by operating a watering device through a servo motor [17-20]. The novelty and major contribution of the proposed work includes :

- (i) Design of a low-maintenance and portable device
- (ii) Automatic operation of a cooling fan, mechanical shade, and water pump.

(iii) Easy access to all parameters across the globe through the IoT platform.

The organizing content of the work includes five subsections where section I discusses the overview of the smart planting monitoring system. In section II the Methodology utilized in this work is elaborated followed by experimental tests in section III. Section IV elaborated results analysis and discussion followed by concluding remarks in section V.

II. METHODOLOGY

The methodology of the proposed project work is explained in terms of techniques and tools applied and technical specification of components. The complete block diagram of the proposed smart plant monitoring and controlling System for greenhouse application is shown in Fig. 1.

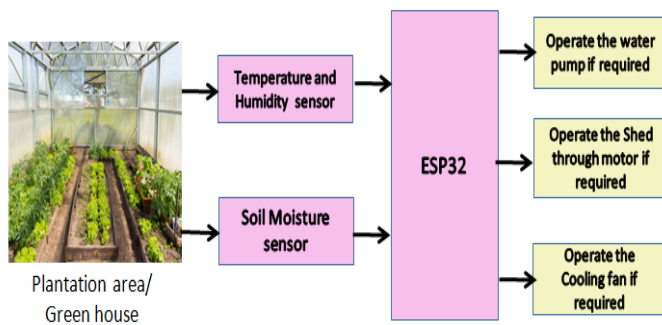


Fig. 1 Block diagram of the proposed plant monitoring and controlling system for greenhouse application

A) Applied Techniques and Tools

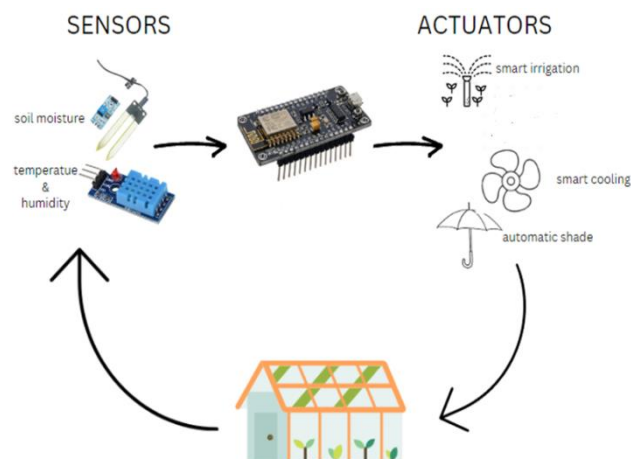


Fig. 2 Smart plant monitoring system

This prototype includes a DHT11 sensor can be mounted within a greenhouse, a soil moisture sensor installed near the plant roots, and an LDR sensor to detect the intensity of light falling on the greenhouse. The two sensors provide their data to the microcontroller, which analyses it. The microcontroller checks the data to the program's specified values, and if the temperature exceeds the specified range, the fan is activated as needed shown in Fig. 2 and Fig. 3. The LDR sensor detects the intensity of the light, and if it is greater than the appropriate level, a shade will be placed over

the structure to minimise the quantity of sunlight falling on the greenhouse, so lowering the temperature. If the soil moisture is determined to be less than the specified level, the microcontroller will activate the motor to pump water into the soil. The soil moisture sensor will continue to measure the moisture level, and when it is sufficient, the microcontroller will turn off the motor automatically. All of this would be linked to a mobile application through a Wi-Fi/Bluetooth module, which would convey all essential data to the user.

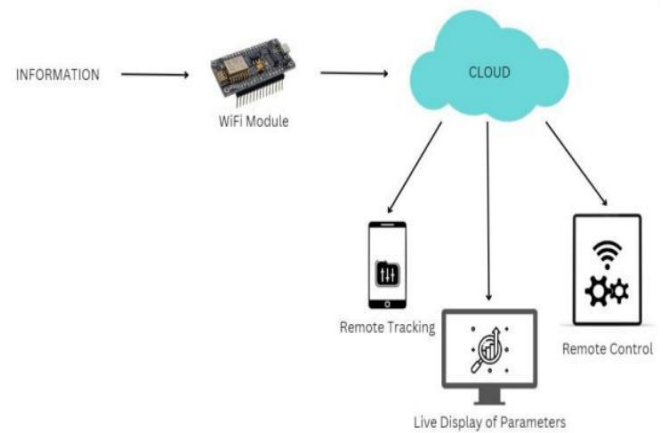


Fig. 3 Functional diagram of prototype

B) Technical Specifications

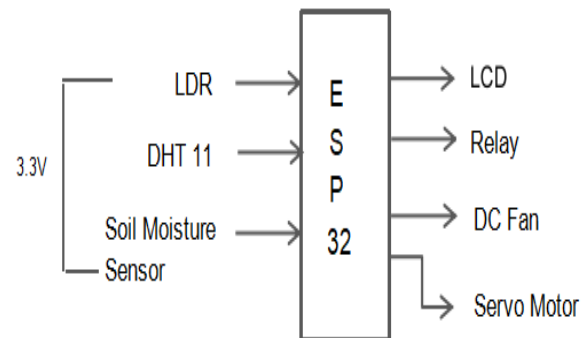


Fig. 3 Illustration of major input and output pins of the microcontroller

Our primary microprocessor board is an ESP32. An LDR sensor, a DHT11 (humidity and temperature sensor), and a soil moisture sensor are attached to the ESP32's input port. On the other side, the ESP32's output pins are linked to an LCD, a relay with a DC motor attached, a DC fan, and a servo motor shown in Fig. 4.

To measure voltage, an LDR is linked in series with a fixed resistor that is connected to the ESP32. When high-intensity light rays strike the LDR, the resistance of the LDR increases, increasing the voltage of the fixed resistance, which is detected by the ESP32 and activates the servo motor, bringing on the mechanical shade.

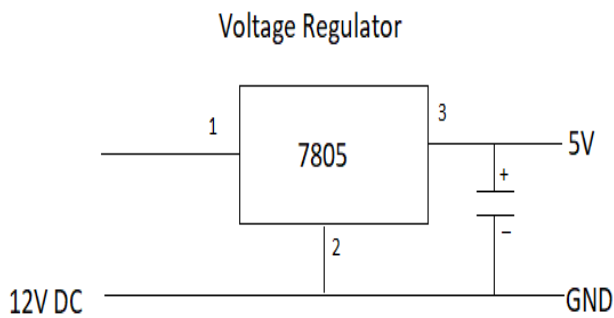


Fig. 5 Schematic of Voltage regulator

Additionally, a voltage regulator (7805) and a filter capacitor are linked to an external power source to operate the cooling fan and water pump, as illustrated in Fig.4 above.

There are four pins in the soil moisture sensor: VCC, Ground, A0 (analog pin), and D0 (digital pin). The Analog pin of the Soil Moisture sensor is linked to the ESP32 to examine the moisture level is illustrated in Fig.6. The digital pin of the soil moisture sensor is not used as we are not interested discrete data.

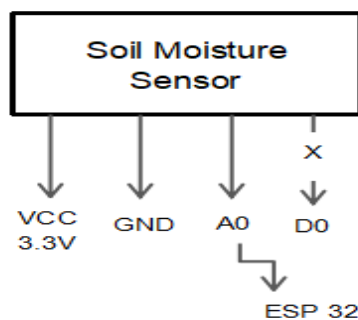


Fig. 6 Schematic of soil moisture sensor

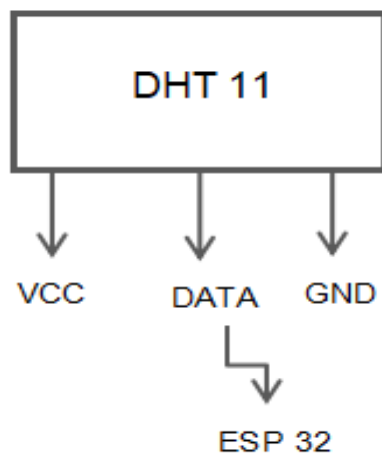


Fig.7 Schematic of DHT 11 sensor

The VCC, Data, and Ground pins are the three that make up the DHT11 sensor. To monitor humidity and temperature, the DHT11 data pin is directly attached to the ESP32 is illustrated in Fig. 7.

A I2C module connects the LCD to the ESP32. VCC,GND,SDA (serial data), and SCL (serial clock) are the four pins on the I2C module. The ESP32's SDA and SCL pins are coupled to those of the I2C module is illustrated in Fig.8.

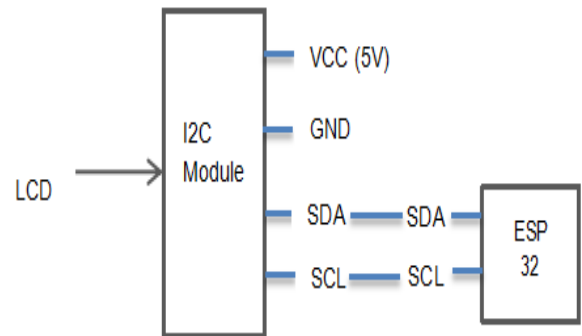


Fig. 8 Schematic for LCD

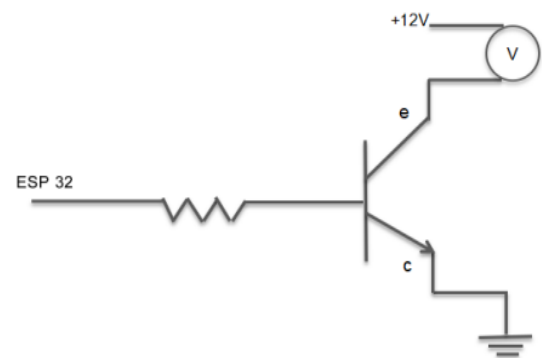


Fig. 9 Schematic diagram of fan motor

A signal is supplied directly from the ESP32 to the cooling fan via a transistor. The cooling fan is charged with 12 volts through the external power supply illustrated in Fig. 8. The transistor's collector and emitter create a channel when the ESP32 delivers a "HIGH" (3.3 volts) signal, turning it on and turning on the DC fan in the process.

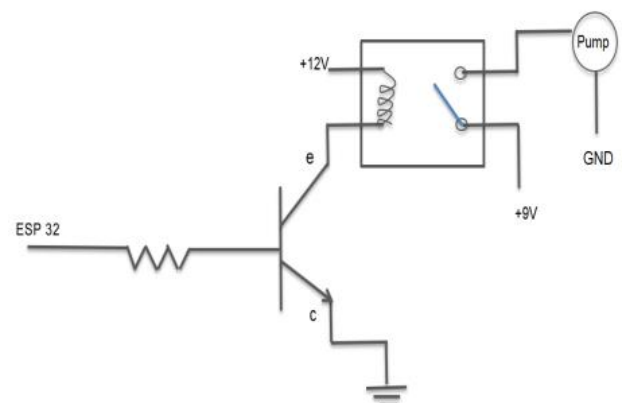


Fig. 10 Schematic of DC water pump

An external 9 volt battery is linked to the DC pump illustrated in Fig. 10. Because the DC pump uses a lot of current, a relay is also linked to the ESP32 through a transistor. When the ESP32 delivers a 'HIGH' (3.3v) signal, the transistor turns on, which turns on the relay and the water pump.

III. EXPERIMENTS AND TESTS

We started with a basis from which the entire arrangement will be created. A box for the plants, a water tumbler, and some room for the circuit are all on it. The project's sensors include soil moisture sensors, temperature sensors, and humidity sensors. The project's actuators include a water pump, a servomotor for the mechanical shade, and a cooling fan. In addition, a battery, a relay, and a WiFi module ESP32 are employed in the complete prototype is illustrated in Fig. 11.

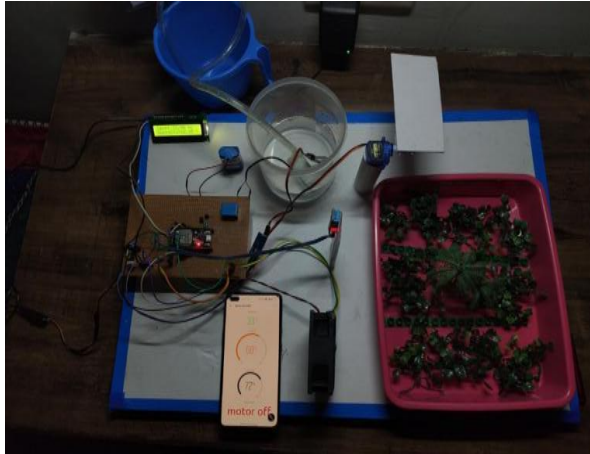


Fig.11 The complete prototype

IV. RESULT ANALYSIS AND DISCUSSION

The designed prototype is built as per the circuit and the simulations are carried out using Blynk App. The temperature, humidity, soil moisture, and water pump status are all correctly displayed by the Blynk App as illustrated in Fig. 12, which operates flawlessly and provides accurate readings for all of these variables. When the temperature exceeds 35 degrees Celsius, the cooling fan is activated as expected and remains turned on until the temperature falls below 35 degrees Celsius. When soil moisture falls below 30%, our water pump is activated as illustrated in Fig. 14.



Fig.12 The Blynk App. IoT platform

When the LDR detects a high intensity of light, the connected shade will be automatically deployed to block direct sunlight. The prototype with and without the mechanical shade is given in Fig. 13 and Fig. 14 respectively. The working of the motor pump is shown in Fig. 15 as the motor is on display on the mobile device.



Fig. 13 Prototype without shade



Fig.14 Prototype with mechanical shade

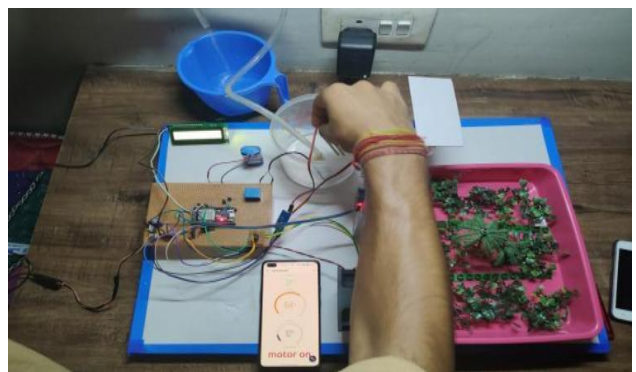


Fig.15 Working of water pump for moisture

V. CONCLUSION

In this work both the hardware and software implementations of the intelligent plant monitoring system is carried out. This prototype provides a plethora of benefits. The inclusion of three intelligent control systems, including a cooling/heating system, an irrigation control system, and a light control system, provides an added advantage to the project. This independent configuration eliminates the requirement for external processing, making our system useful

and self-contained. Our intelligent plant monitoring usage of IoT (Internet of Things) technology has several important benefits. Users may remotely monitor their plants and make decisions based on real-time data using the IoT interface. This function significantly simplifies farmers' jobs and helps them to successfully regulate their greenhouse environment. The sensors and actuators are operating flawlessly and satisfying all the criteria, as we have seen during our testing phase. The Blynk IoT interface provides precise and trustworthy readings, providing consumers with real-time information about their greenhouse. This prototype can provide farmers with an inexpensive, effective, reliable, and user-friendly solution for optimal plant development.

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